SHBT 261 Project 1 - Image Classification on Caltech-101 Dataset

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Code Repository: https://github.com/aalice-z/SHBT-261-mini-project-1.git

1. Introduction

This project aims to (1) compare classical and modern approaches by systematically evaluating three classical ML methods (SVM, Random Forest, KNN) against two deep learning models (ResNet and EfficientNet). (2) Conduct ablation studies to investigate the impact of removing, replacing, disabling and changing model designs. (3) Analyze model performance by examining per-class accuracy patterns to identify categories where different models' strength and challenges. (4) Provide actionable recommendations for model selection and hyperparameter tuning in model training.

This study uses the Caltech-101 Dataset with a total of approximately 9000 images, 101 object categories with approximately 40 to 800 images per category. Images vary in size and RGB color. In terms of data splitting, 70% of the image data is diverted to training, 15% for validation and 15% for testing. Input images are preprocessed, they are resized to specified dimensions (64×64, 128×128 and 224×224 depending on model input requirement) and normalized for deep learning models.

2. Methods

2.1 Classical Machine Learning Approaches

Histogram of Oriented Gradients (HOG) features were extracted, and they capture edge orientation information by computing gradients in localized regions of an image. Principal Component Analysis (PCA) was applied to reduce the dimension and computational cost, mitigating the curse of dimensionality

(A) **Support Vector Machine (SVM)** with Radial Basis Function kernel can model non-linear decision boundaries, suitable for complex visual patterns. Regularization parameter was 1.0 with probability estimates for top-k accuracy computation.

(B) **Random Forest** provides robustness and handles high-dimensional feature spaces well. In this ensemble approach there were 100 estimators, with a max depth of 20 levels and random state set to 42.

(C) **K-Nearest Neighbors (KNN)** with number of neighbors set to 5 and Euclidean distance metric. This approach is sensitive to feature quality and distance metrics.

2.2 Deep Learning Approaches

2.2.1 ResNet-18 Architecture

The base model was ResNet-18 (He et al., 2016) with ImageNet pretrained weights. 18 layers with residual connections. The final fully connected layer modified for 101 classes. Input sizes were 224×224×3. Residual connections addresses vanishing gradient problems through skip connections that allow gradients to flow directly through the network.

2.2.2 EfficientNet-B0 Architecture

The base model was EfficientNet-B0 (Tan & Le, 2019) with ImageNet pretrained weights. 18 layers with residual connections. The final fully connected layer modified for 101 classes. Input sizes were 224×224×3. EfficientNet achieves best accuracy with fewer parameters through systematic architecture scaling and mobile inverted bottleneck blocks.

2.2.3 Training Details

Configurations are the same for both Resnet and EfficientNet:

- Optimizer: Adam (default, β_1=0.9, β_2=0.999, ε=1e-8) or SGD (momentum=0.9)
- Learning rate: 0.001
- Batch size: 32
- Epochs: 20
- Loss function: Cross-Entropy Loss

2.3 Data Augmentation

Data augmentation was applied only for deep learning models to increase dataset diversity.

- Random horizontal flip: p = 0.5
- Random rotation: +/- 15 degrees
- Color jitter: Brightness +/- 20% and contrast +/- 20%
- Normalized with ImageNet statistics, mean=(0.485, 0.456, 0.406) sd=(0.229, 0.224, 0.225)

2.4 Evaluation Metrics

The primary metrics were (1) accuracy, the overall correct prediction rate. (2) Macro-average precision, the average precision across all classes without weighing. (3) Weighted-average precision, that is averaged by class support. (4) Macro-averaged recall across all classes. (5) Weighted-average recall averaged by class support. (6) Macro F1-score, the harmonic mean of macro precision and recall. (7) Weighted F1-score, averaged by class support. Additionally top 3, top 5 and per-class accuracy were used.

Confusion matrices are used to identify common misclassifications. Per-class accuracy bar plots are used to identify categories that were difficult for the classification models. Training curves plotting loss and accuracy against epochs were generated for deep learning models.

2.5 Ablation Studies

Four systematic ablation studies were done to understand the impact of design choices.

(A) **Data Augmentation Study:** With compared to without augmentation on ResNet-18 model to evaluate the impact on generalization and overfitting.
(B) **Optimizer Study:** Using Adam compared to SGD with momentum on REsNet-18 model. Learning rates were the same for both choices of optimizers.
(C) **Feature Representation Study:** Using HOG features compared to raw pixel features with the SVM classifier.
(D) **Image Size Study:** Comparing size 64×64 to 128×128 input resolution on Random Forest model with HOG feature to evaluate the trade-off between resolution and computational cost.

3. Results

3.1 Overall Model Performance Comparison

Table 1: Performance Metrics for All Models

Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)	Top-3 Acc	Top-5 Acc
EfficientNet-B0	94.52%	92.74%	91.69%	91.71%	98.61%	99.10%
ResNet-18	84.30%	80.28%	75.99%	75.84%	94.19%	96.08%
SVM (HOG)	68.68%	61.77%	51.74%	53.08%	79.89%	84.38%
KNN (HOG)	58.22%	48.74%	37.78%	39.15%	67.70%	71.79%
Random Forest (HOG)	56.66%	47.69%	34.63%	35.88%	67.21%	71.95%

Findings:

1. Deep learning model EfficientNet-B0 outperforms the best classical method (SVM) by 25.84% in accuracy.
2. EfficientNet-B0 exceeds ResNet-18 by 10.22% even with fewer parameters, proving the effectiveness of compound scaling.
3. All classical approaches achieve below 70% accuracy, with Random Forest performing worst (56.66% accuracy).
4. Deep models show exceptional top 3 and top 5 accuracy (the additional performance metrics used), with EfficientNet achieving 99.10% top 5 accuracy.

Figure 1: Model Accuracy Comparison

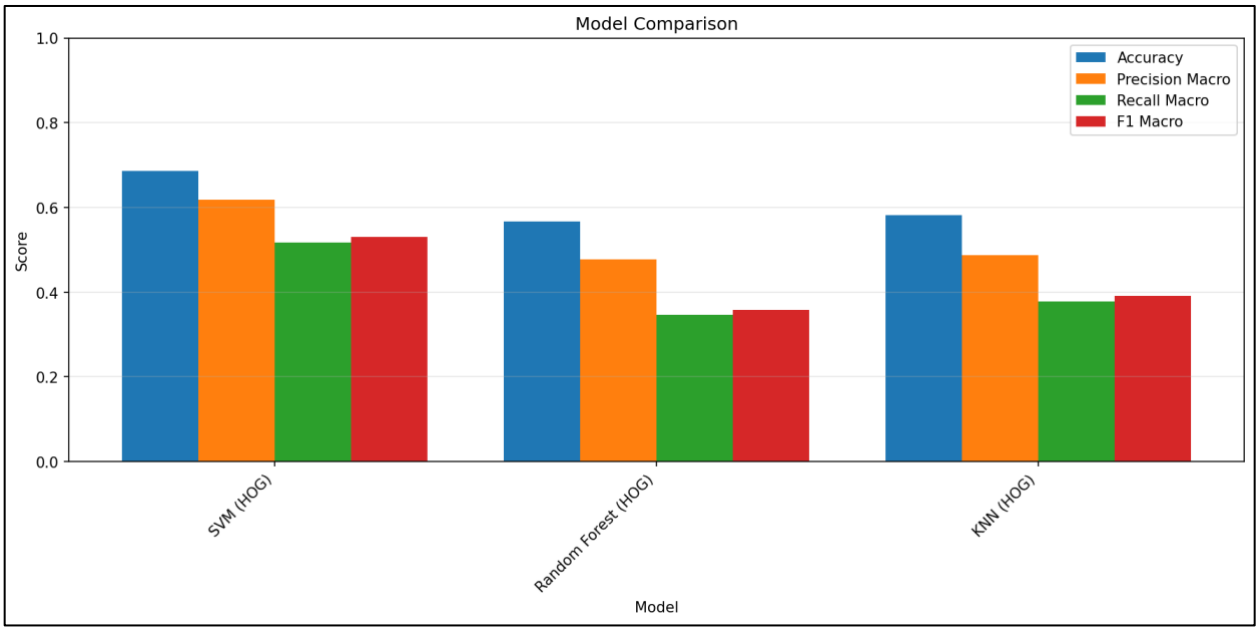

Figure 1: Performance comparison of classical ML models (SVM, Random Forest, KNN) on Caltech-101 dataset using HOG features. SVM achieves the best performance among classical approaches with 68.68% accuracy.

3.2 Deep Learning Model Analysis

3.2.1 ResNet-18 Performance

Table 2: ResNet-18 Detailed Metrics

Metric	Value
Test Accuracy	84.30%
Precision (Weighted)	86.65%
Recall (Weighted)	84.30%
F1-Score (Weighted)	84.10%
Top-3 Accuracy	94.19%
Top-5 Accuracy	96.08%

Figure 2: Training Dynamics

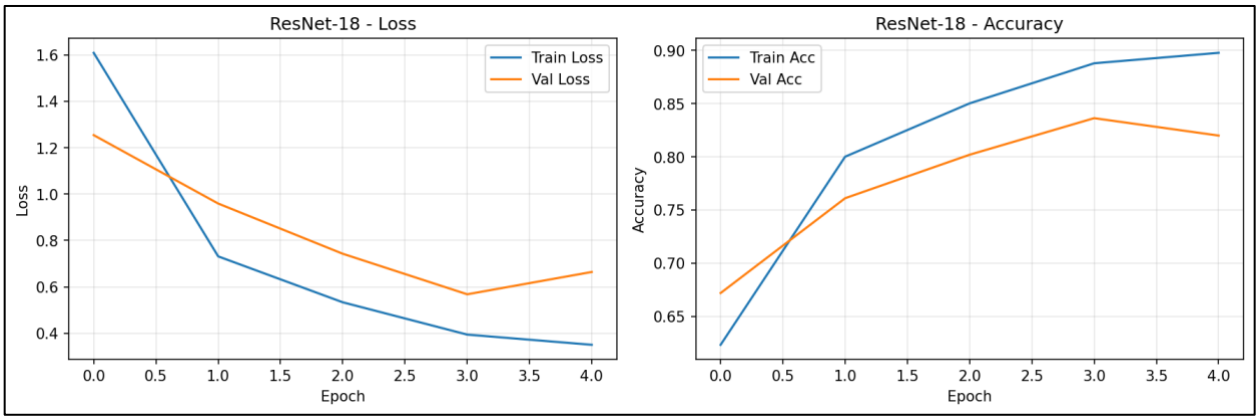

Figure 2: Training and validation curves for ResNet-18. Left: Cross-entropy loss over 20 epochs showing convergence. Right: Accuracy curves demonstrating minimal overfitting with final validation accuracy of ~84%.

Per-Class Performance:

1. Perfect accuracy (100%): Faces, Leopards, Motorbikes, accordion, brain, dollar_bill
2. Strong performance (>90%): airplanes (98.3%), car_side (94.7%)
3. Challenging (<40%): anchor (16.7%), ant (16.7%), chair (40%), crocodile_head (14.3%)

Figure 3: ResNet-18 Confusion Matrix

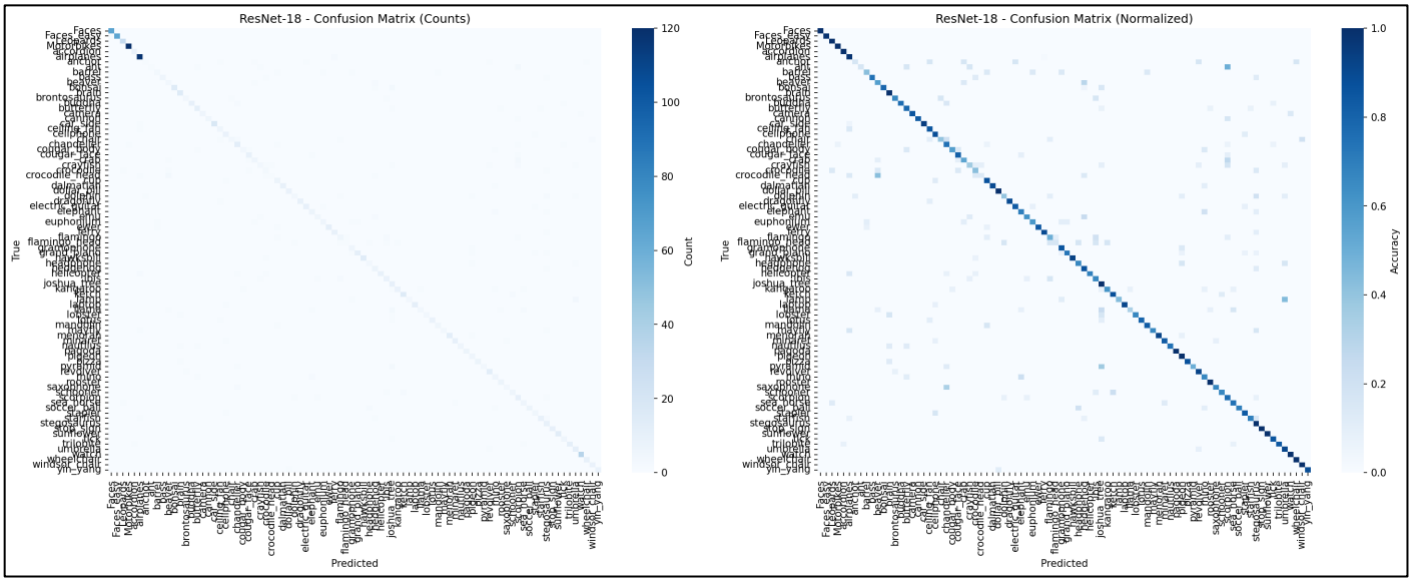

Figure 3: Confusion matrix for ResNet-18 on test set. Diagonal elements show correct classifications. Off-diagonal elements indicate common confusions, particularly among visually similar categories.

Figure 4: ResNet-18 Per-Class Accuracy

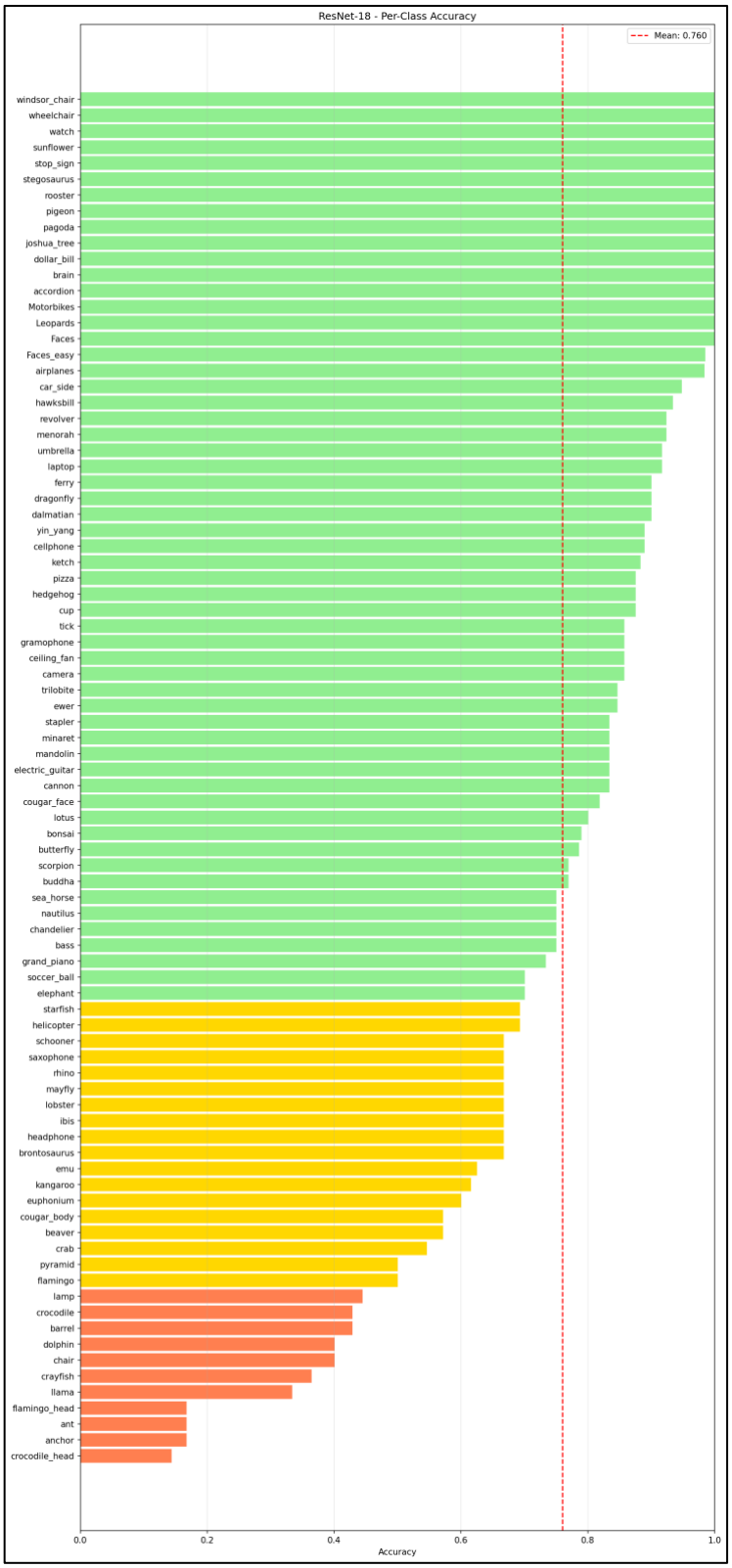

Figure 4: Per-class accuracy for ResNet-18 across all 101 categories. Wide variance illustrates dataset imbalance and inherent class difficulty.

3.2.2 EfficientNet-B0 Performance

Table 3: EfficientNet-B0 Detailed Metrics

Metric	Value
Test Accuracy	94.52%
Precision (Weighted)	95.03%
Recall (Weighted)	94.52%
F1-Score (Weighted)	94.48%
Top-3 Accuracy	98.61%
Top-5 Accuracy	99.10%

Figure 5: Training Dynamics

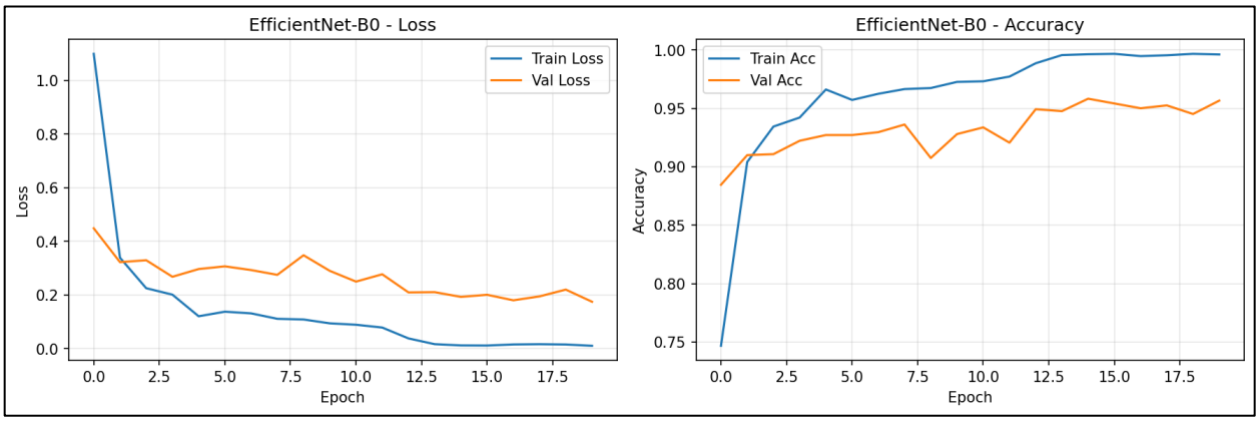

Figure 5: Training and validation curves for EfficientNet-B0. Smoother convergence compared to ResNet-18 with higher final accuracy (around 94%) and minimal overfitting gap.

Per-Class Performance:

1. Perfect accuracy (100%): Leopards, Motorbikes, accordion, bonsai, brain, buddha, camera, cannon, car_side, cellphone, chandelier, cup, dalmatian, dollar_bill, dragonfly, elephant, euphonium
2. Strong performance (>90%): airplanes (99.2%), cougar_face (90.9%), butterfly (92.9%)
3. Some challenging categories: anchor (50%), brontosaurus (50%), crayfish (54.5%), crocodile_head (57.1%)

EfficientNet-B0 achieves perfect or near-perfect accuracy on 17 classes, compared to 6 for ResNet-18, demonstrating superior feature learning capabilities.

Figure 6: EfficientNet-B0 Confusion Matrix

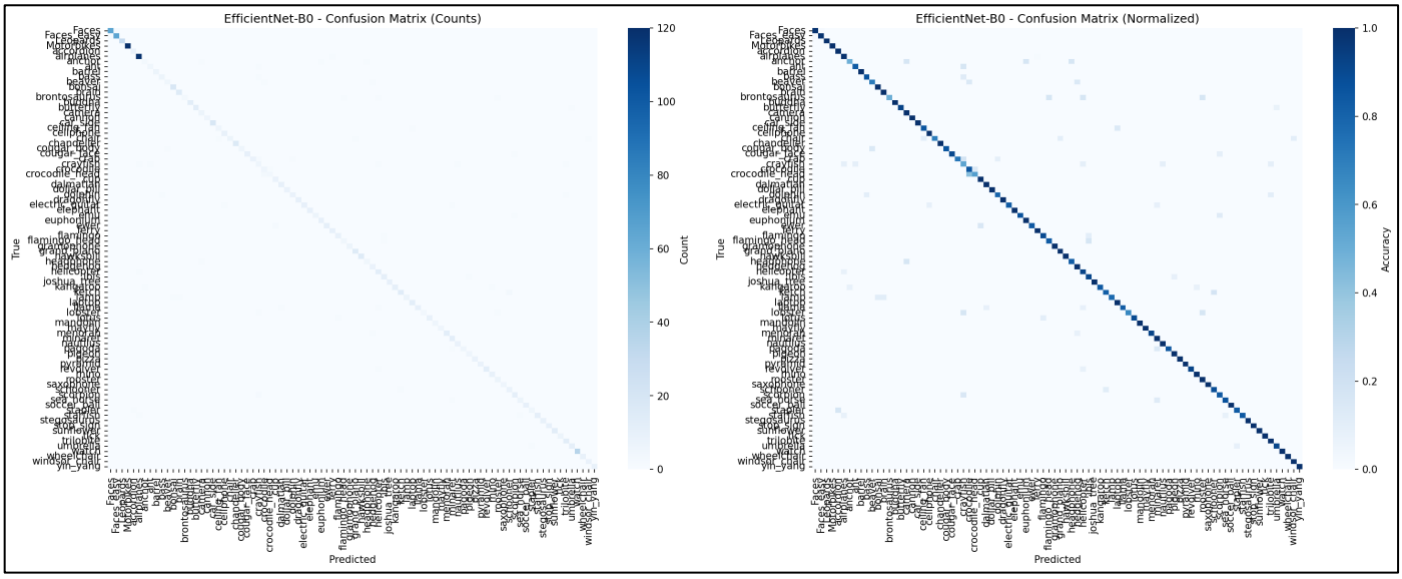

Figure 6: Confusion matrix for EfficientNet-B0. Significantly cleaner diagonal compared to ResNet-18, indicating fewer misclassifications.

Figure 7: EfficientNet-B0 Per-Class Accuracy

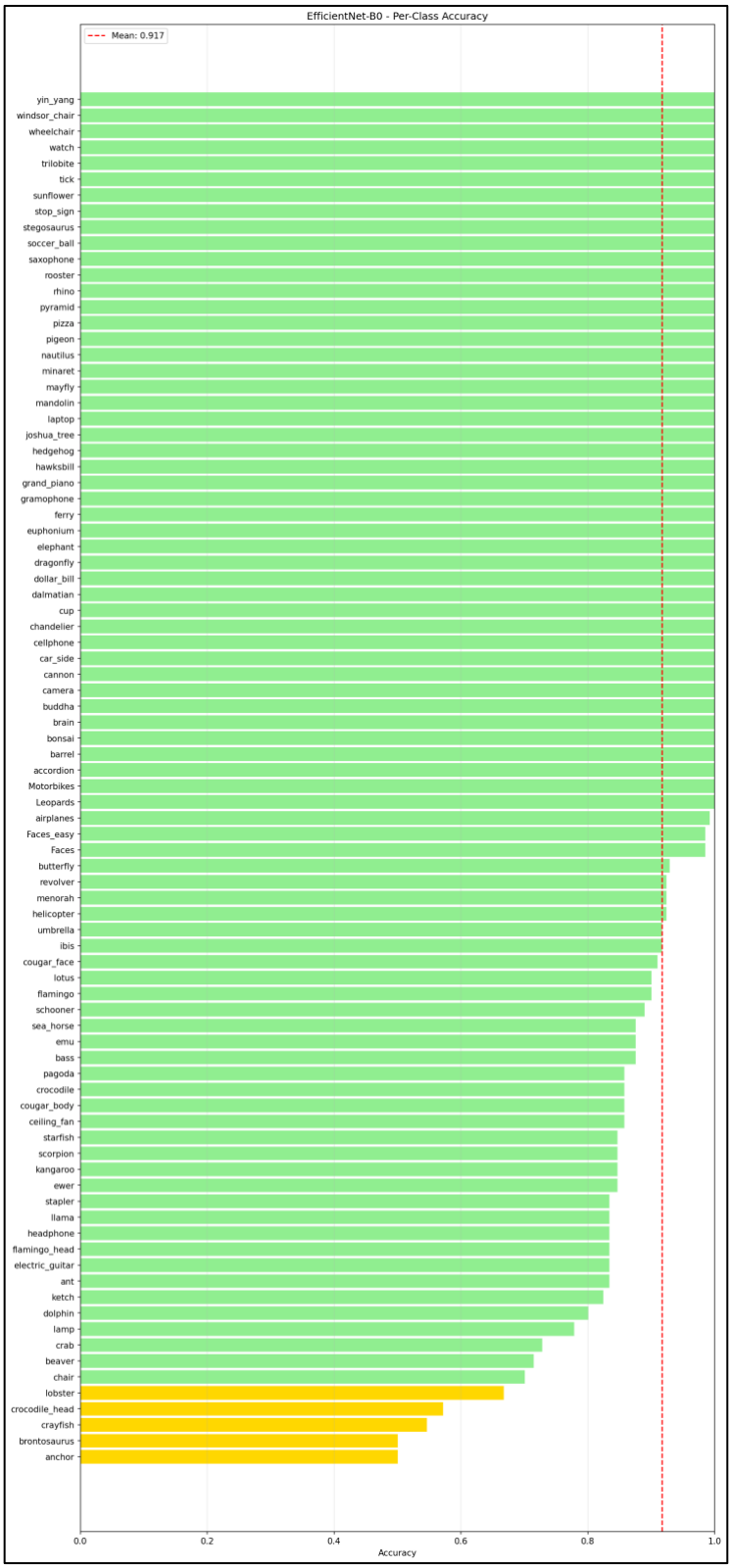

Figure 7: Per-class accuracy for EfficientNet-B0. Most classes exceed 80% accuracy with fewer problematic categories compared to other models.

3.3 Classical Model Analysis

3.3.1 Support Vector Machine (SVM)

Table 4: SVM with HOG Features Performance

Metric	Value
Test Accuracy	68.68%
Precision (Macro)	61.77%
Recall (Macro)	51.74%
F1-Score (Macro)	53.08%

1. Best performing classical method
2. Strong performance on face categories (Faces: 98.5%, Faces_easy: 98.5%)
3. Perfect accuracy on Leopards, Motorbikes, accordion, airplanes
4. Struggles with fine-grained distinctions (bass: 0%, ceiling_fan: 14.3%, cannon: 16.7%)

Figure 8: SVM Confusion Matrix

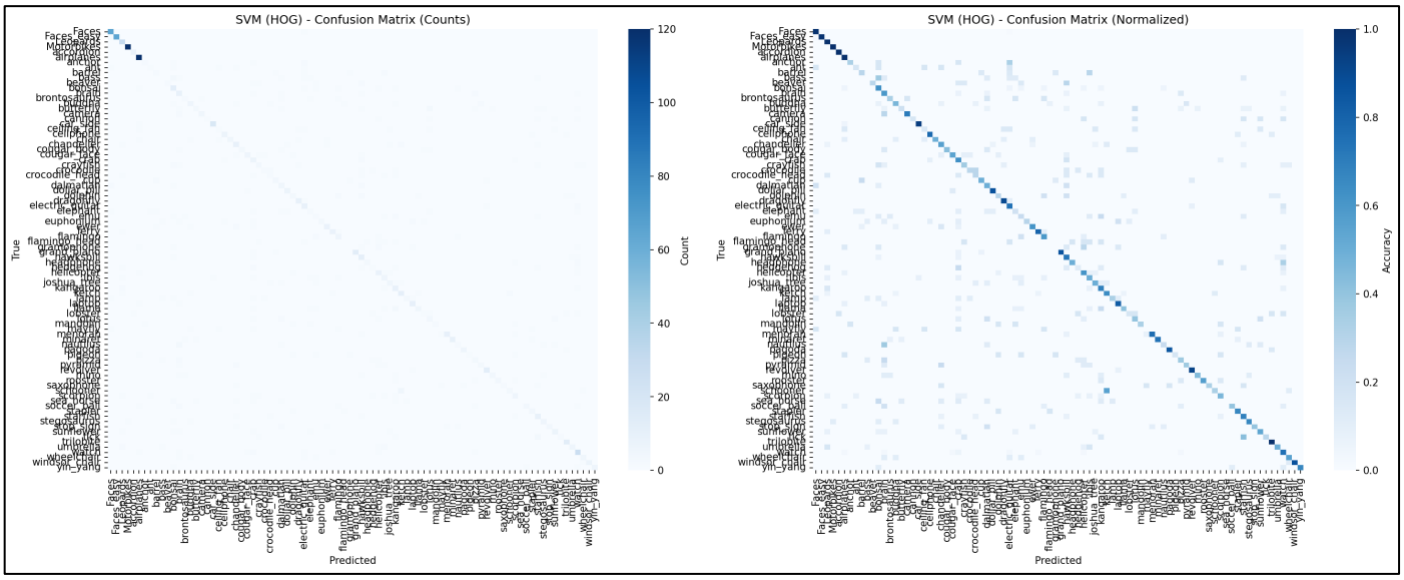

Figure 8: Confusion matrix for SVM with HOG features. Shows reasonable discrimination for some categories but significant confusion for others.

Figure 9: SVM Per-Class Accuracy

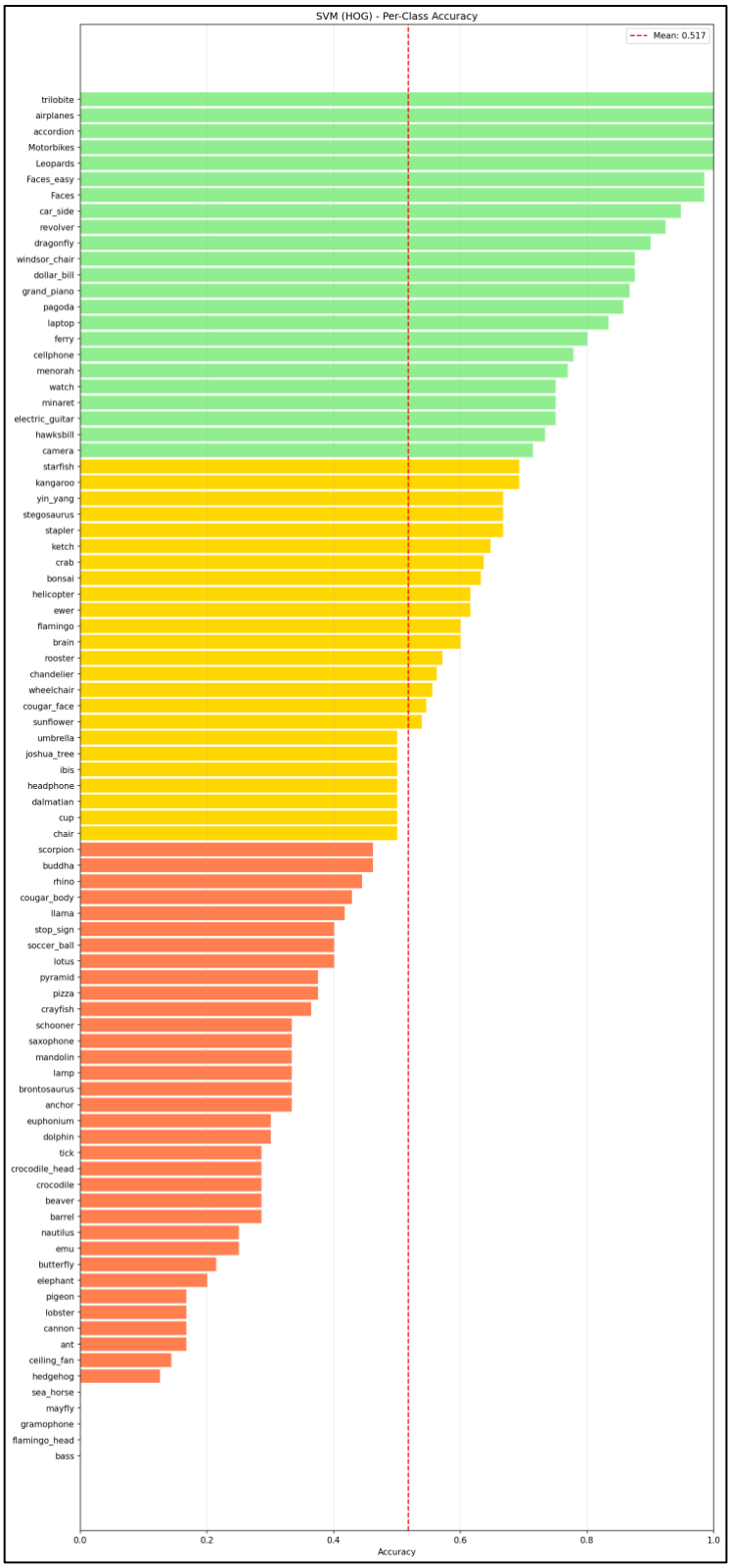

Figure 9: Per-class accuracy for SVM. High variance across classes with several categories showing near-zero accuracy.

3.3.2 Random Forest

Table 5: Random Forest with HOG Features Performance

Metric	Value
Test Accuracy	56.66%
Precision (Macro)	47.69%
Recall (Macro)	34.63%
F1-Score (Macro)	35.88%

1. Ensemble method underperforms individual SVM
2. Excellent on easy categories (Faces: 98.5%, airplanes: 99.2%)
3. Zero accuracy on multiple classes (ant, bass, cannon, chair, cougar_body, crocodile, dalmatian, dolphin, emu)

Figure 10: Random Forest Confusion Matrix

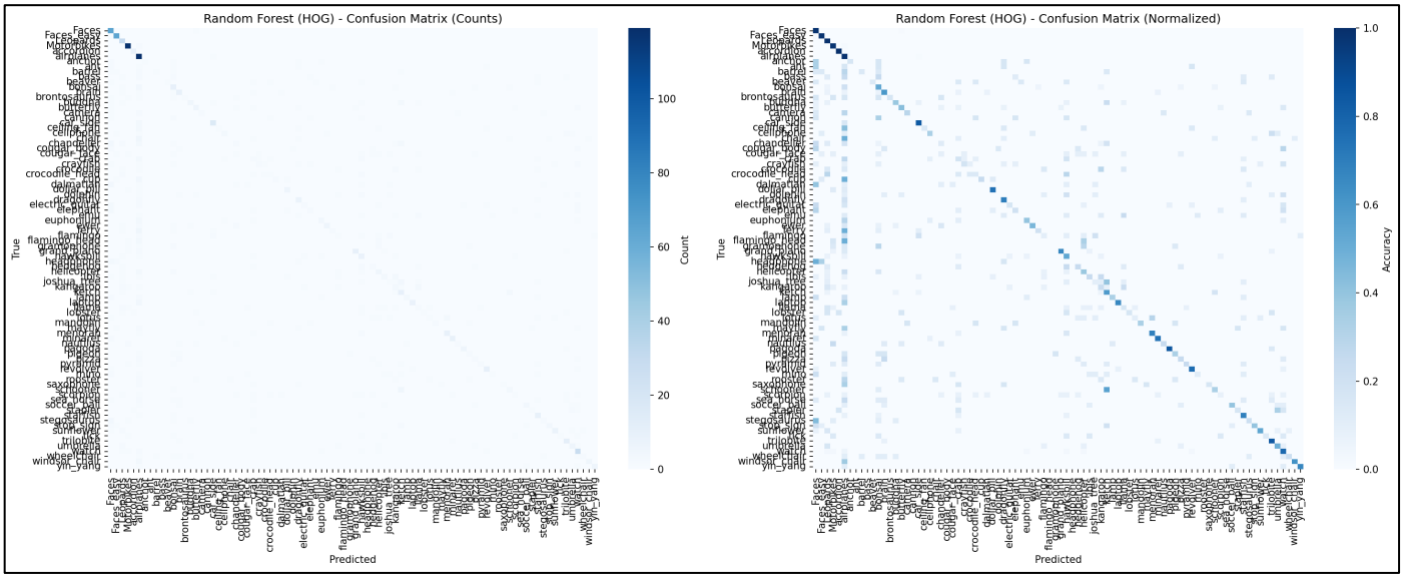

Figure 10: Confusion matrix for Random Forest. More diffuse predictions indicate overfitting to training data or insufficient discrimination.

Figure 11: Random Forest Per-Class Accuracy

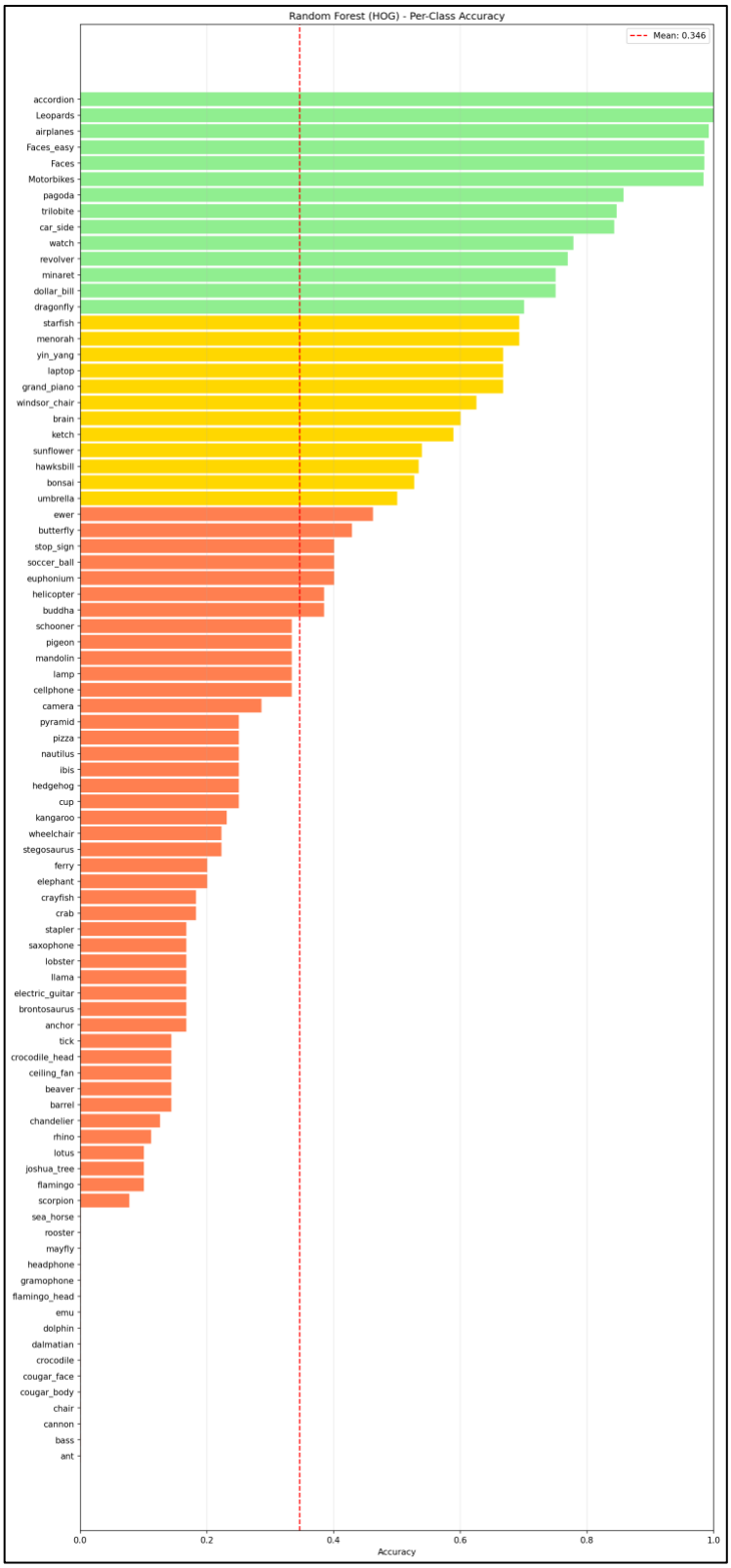

Figure 11: Per-class accuracy for Random Forest. Many classes with zero accuracy suggest poor generalization.

3.3.3 K-Nearest Neighbors (KNN)

Table 6: KNN with HOG Features Performance

Metric	**Value**
Test Accuracy	58.22%
Precision (Macro)	48.74%
Recall (Macro)	37.78%
F1-Score (Macro)	39.15%

1. Slightly outperforms Random Forest
2. Instance-based learning sensitive to feature quality
3. Similar pattern of high variance in per-class accuracy

Figure 12: KNN Confusion Matrix

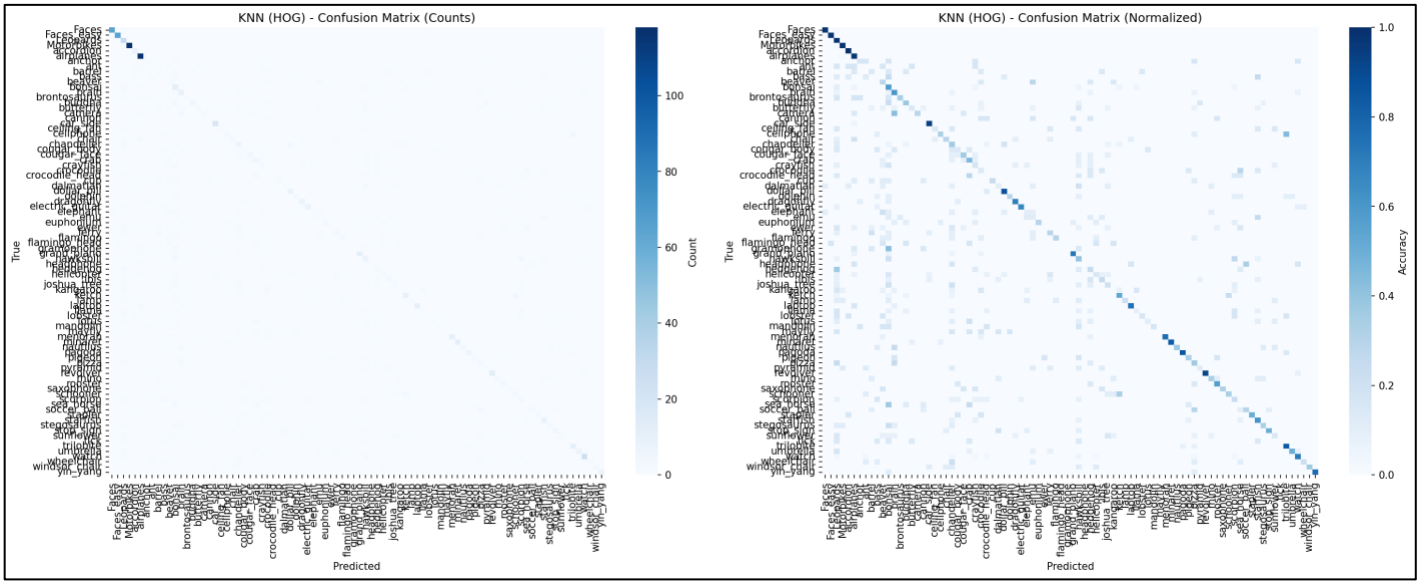

Figure 12: Confusion matrix for KNN (k=5). Non-parametric approach shows limitations with HOG features in high-dimensional space.

Figure 13: KNN Per-Class Accuracy

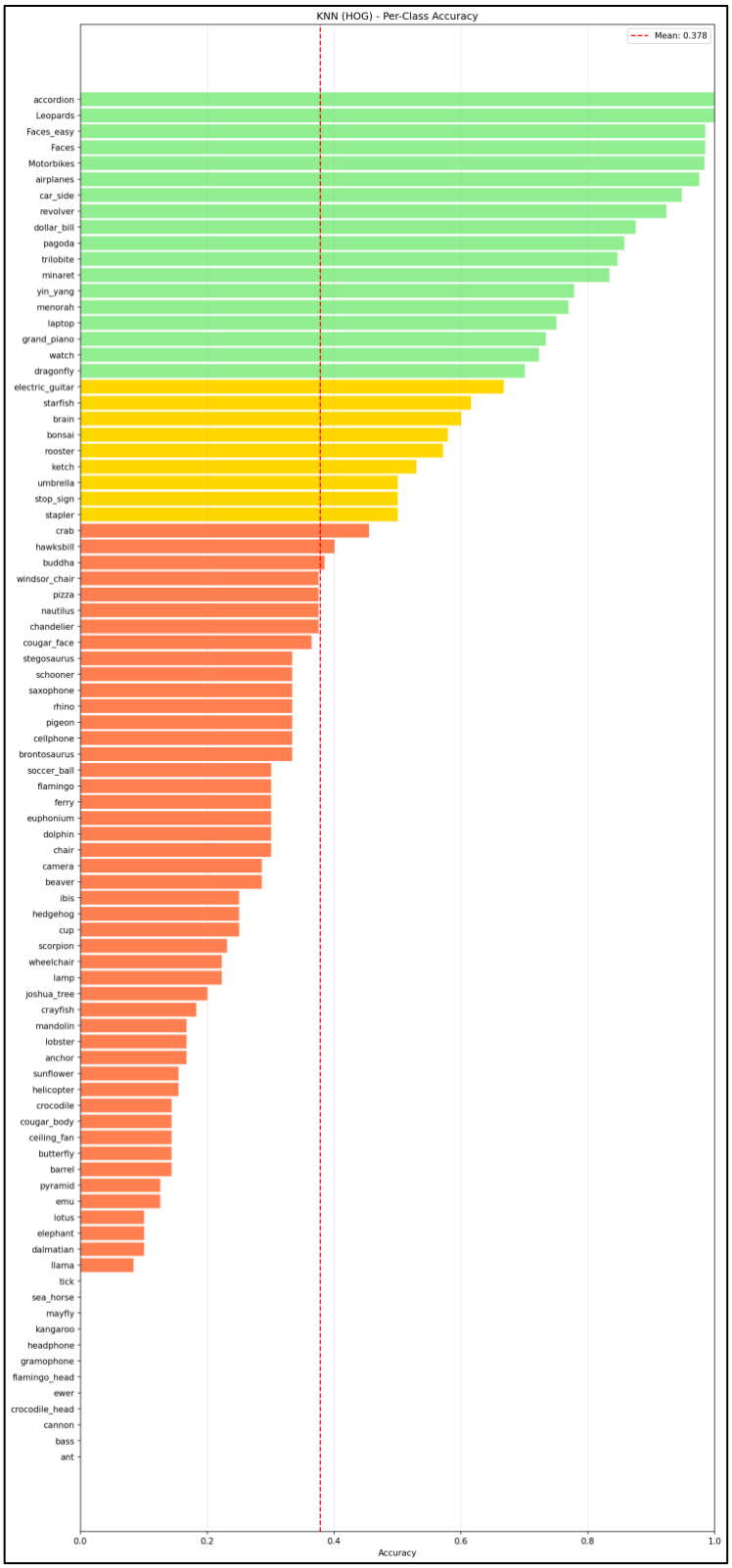

Figure 13: Per-class accuracy for KNN. Performance pattern similar to Random Forest with high variance.

3.4 Ablation Study Results

3.4.1 Data Augmentation Impact

Table 7: Ablation Study - Data Augmentation (ResNet-18)

Configuration	Accuracy	F1-Macro	Precision-Macro	Recall-Macro
With Augmentation	85.69%	78.45%	83.40%	78.57%
Without Augmentation	84.30%	75.84%	80.28%	75.99%

1. Data augmentation provides modest but consistent improvements across all metrics
2. Particularly beneficial for macro-averaged metrics, suggesting better performance on difficult classes
3. Reduced overfitting (not shown in table but evident from validation curves)

Figure 14: Data Augmentation Ablation

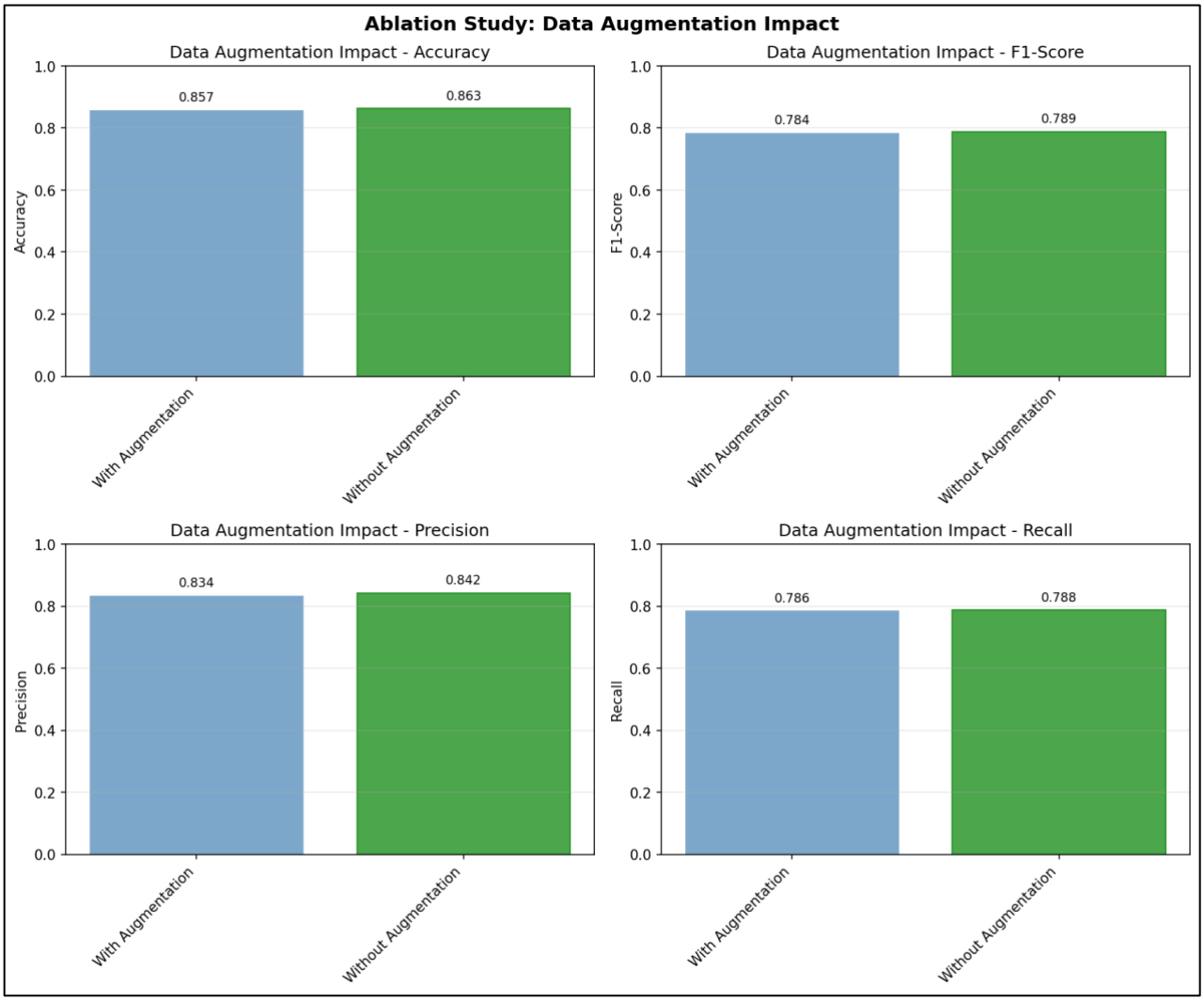

Figure 14: Impact of data augmentation on ResNet-18 performance. All metrics improve with augmentation, particularly macro-averaged scores.

Figure 15: Augmentation Confusion Matrices

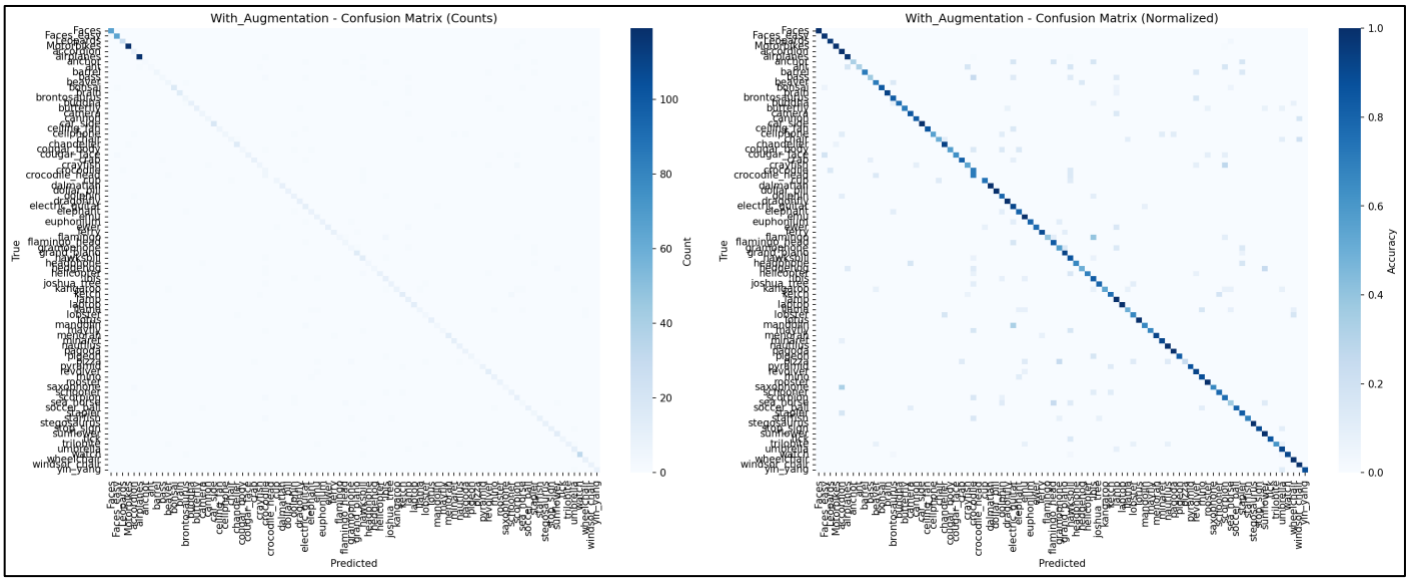

Figure 16: Without Augmentation Confusion Matrix

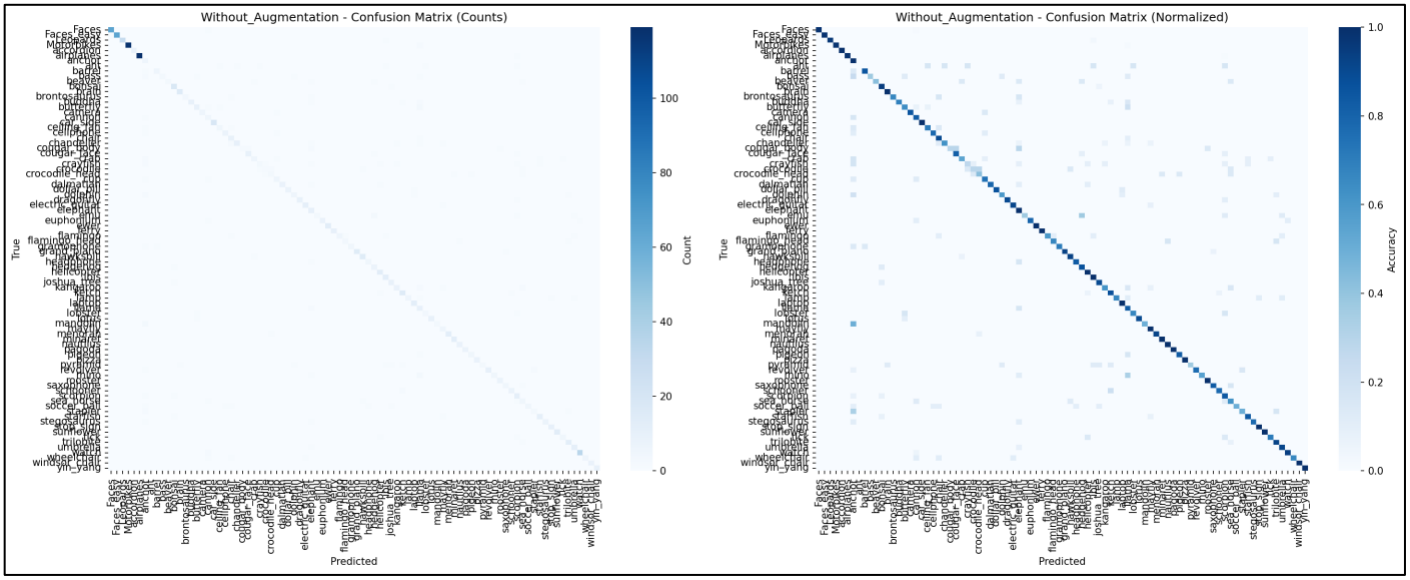

Figure 16: Confusion matrix for ResNet-18 without data augmentation. Slightly more off-diagonal errors compared to augmented version.

3.4.2 Optimizer Comparison

Table 8: Ablation Study - Optimizer Choice (ResNet-18)

Optimizer	Accuracy	F1-Macro	Precision-Macro	Recall-Macro
Adam	87.65%	81.07%	85.18%	81.32%
SGD (momentum=0.9)	84.30%	75.84%	80.28%	75.99%

1. Adam optimizer significantly outperforms SGD with momentum
2. Larger improvements in macro-averaged metrics (+5.23% F1-macro) suggest Adam handles difficult classes better
3. Adam's adaptive learning rates likely more suitable for this moderately-sized dataset

Figure 17: Optimizer Ablation

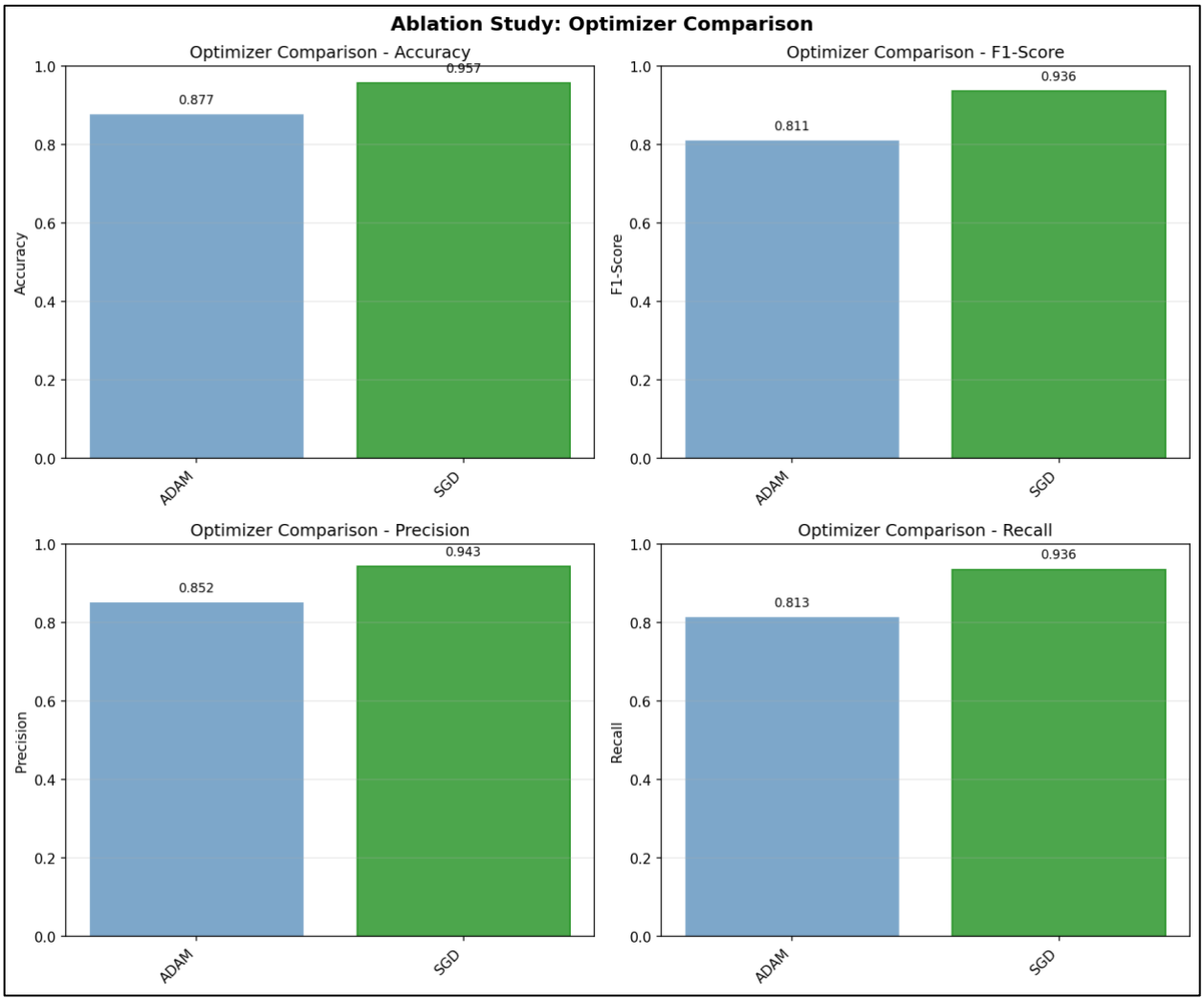

Figure 17: Optimizer comparison for ResNet-18. Adam consistently outperforms SGD across all metrics.

Figure 18: Adam Confusion Matrix

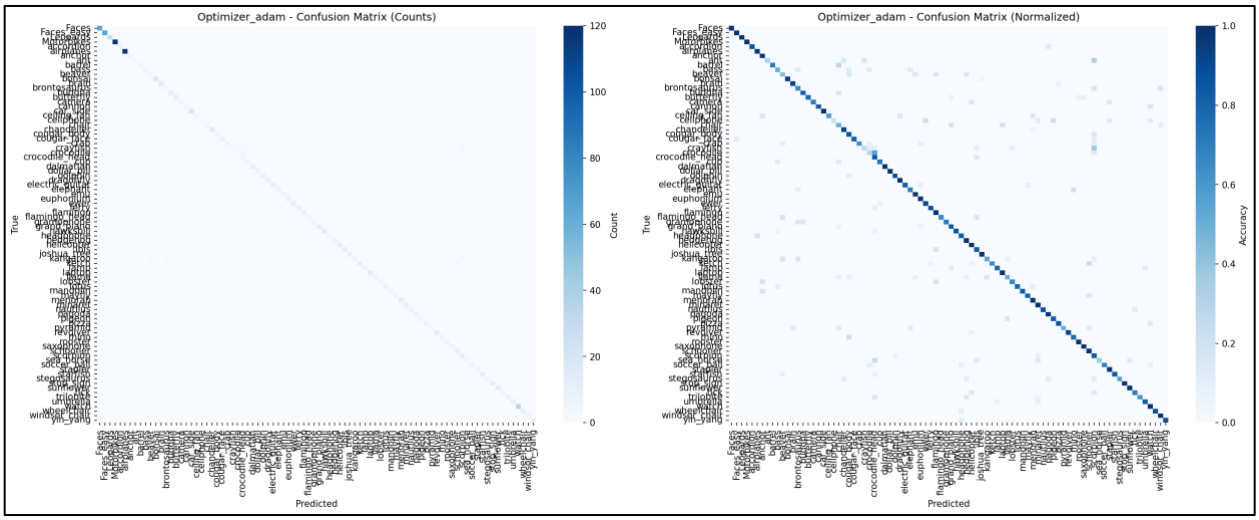

Figure 18: Confusion matrix for ResNet-18 with Adam optimizer. Cleaner diagonal indicates better overall classification.

Figure 19: SGD Confusion Matrix

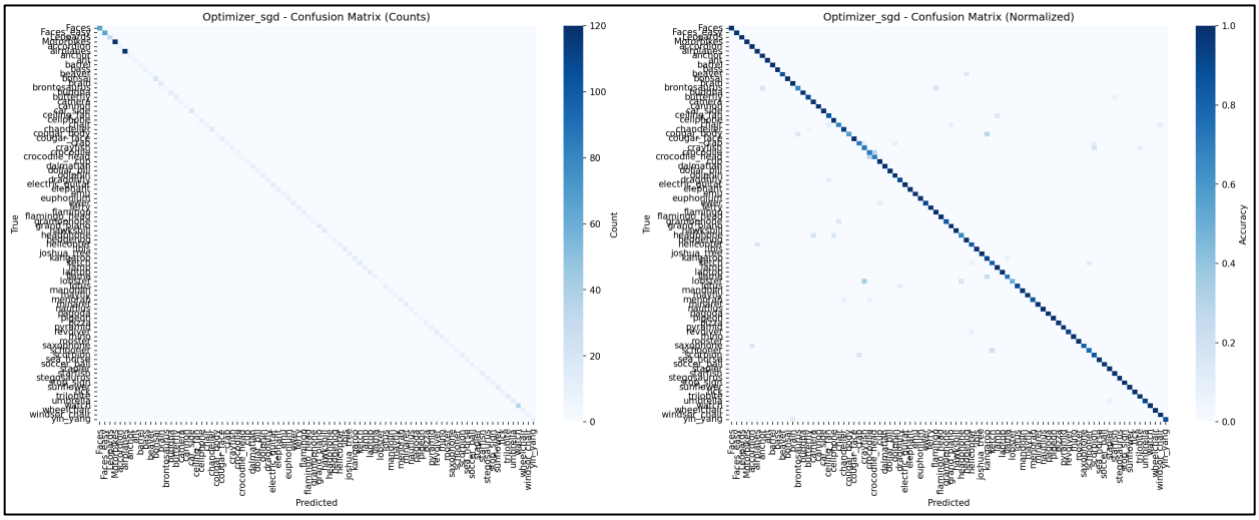

Figure 19: Confusion matrix for ResNet-18 with SGD optimizer. More confusion compared to Adam.

3.4.3 Feature Representation Comparison

Table 9: Ablation Study - Feature Type (SVM)

Feature Type	Accuracy	F1-Macro	Precision-Macro	Recall-Macro
HOG	58.46%	39.35%	51.71%	37.73%
Raw Pixels	49.42%	32.18%	43.82%	28.95%

1. Hand-crafted HOG features substantially outperform raw pixels (+9.04% accuracy)
2. HOG's orientation histograms capture relevant shape information ignored by raw intensities
3. Demonstrates importance of feature engineering in classical ML approaches
4. Raw pixels suffer from high dimensionality and sensitivity to pixel-level variations

Figure 20: Feature Ablation

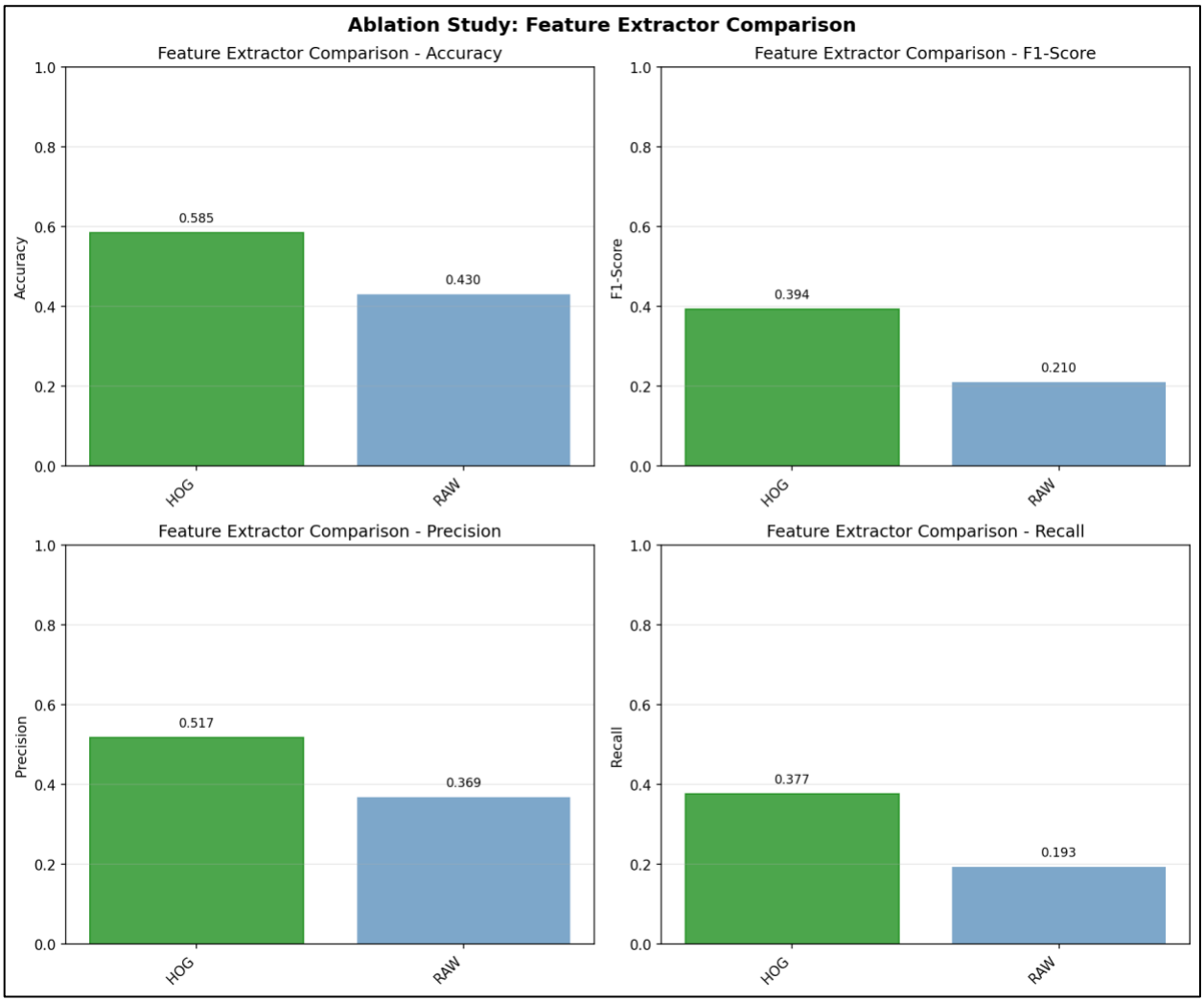

Figure 20: Comparison of HOG vs. raw pixel features using SVM classifier. HOG features provide substantial improvement.

Figure 21: HOG Feature Confusion Matrix

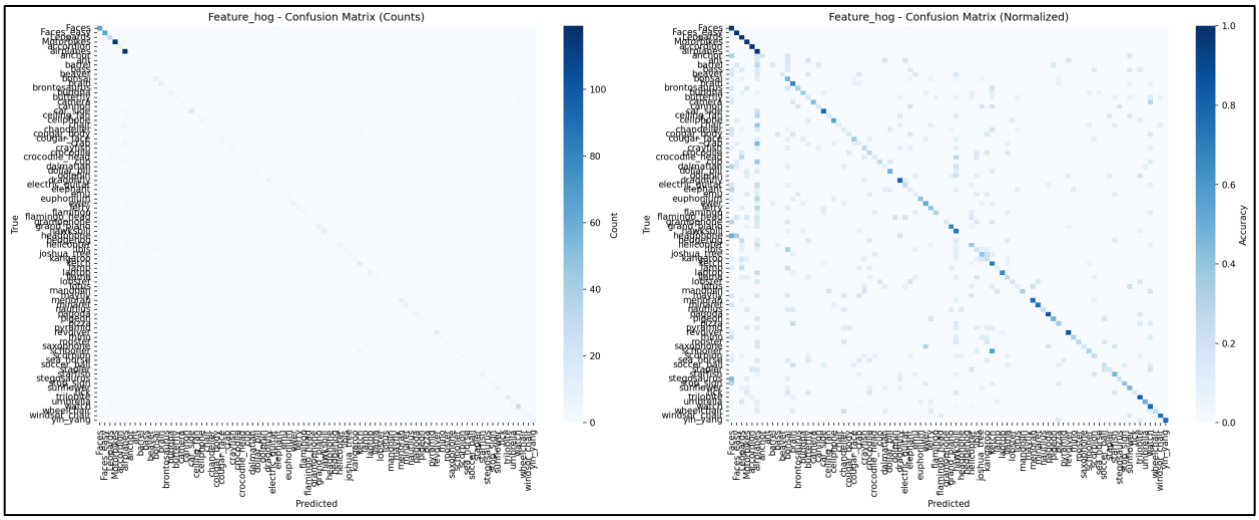

Figure 21: Confusion matrix for SVM with HOG features.

Figure 22: Raw Feature Confusion Matrix

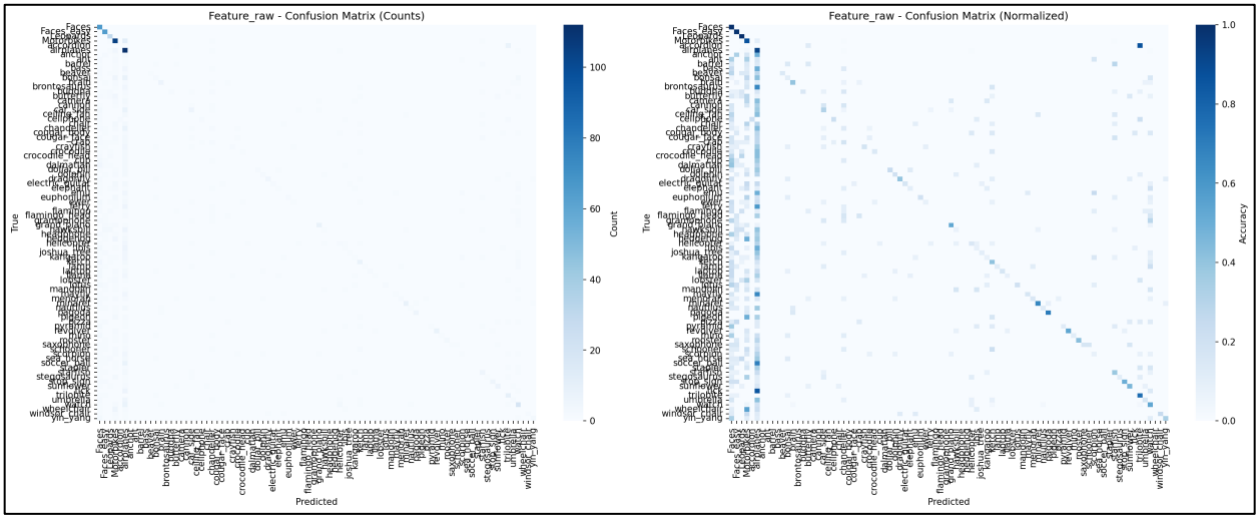

Figure 22: Confusion matrix for SVM with raw pixel features. More dispersed errors indicate poorer discrimination.

3.4.4 Image Size Impact

Table 10: Ablation Study - Input Resolution (Random Forest + HOG)

Image Size	Accuracy	F1-Macro	Precision-Macro	Recall-Macro
128×128	58.46%	39.35%	51.71%	37.73%
64×64	56.17%	34.99%	49.07%	33.35%

1. Higher resolution (128×128) consistently outperforms lower resolution (64×64)
2. Larger improvements in recall-based metrics suggest better detection of difficult classes
3. At 64×64, fine-grained details are lost, particularly affecting texture-dependent categories
4. Trade-off: 128×128 requires approvimately 4× more computation but provides meaningful accuracy gains

Figure 23: Image Size Ablation

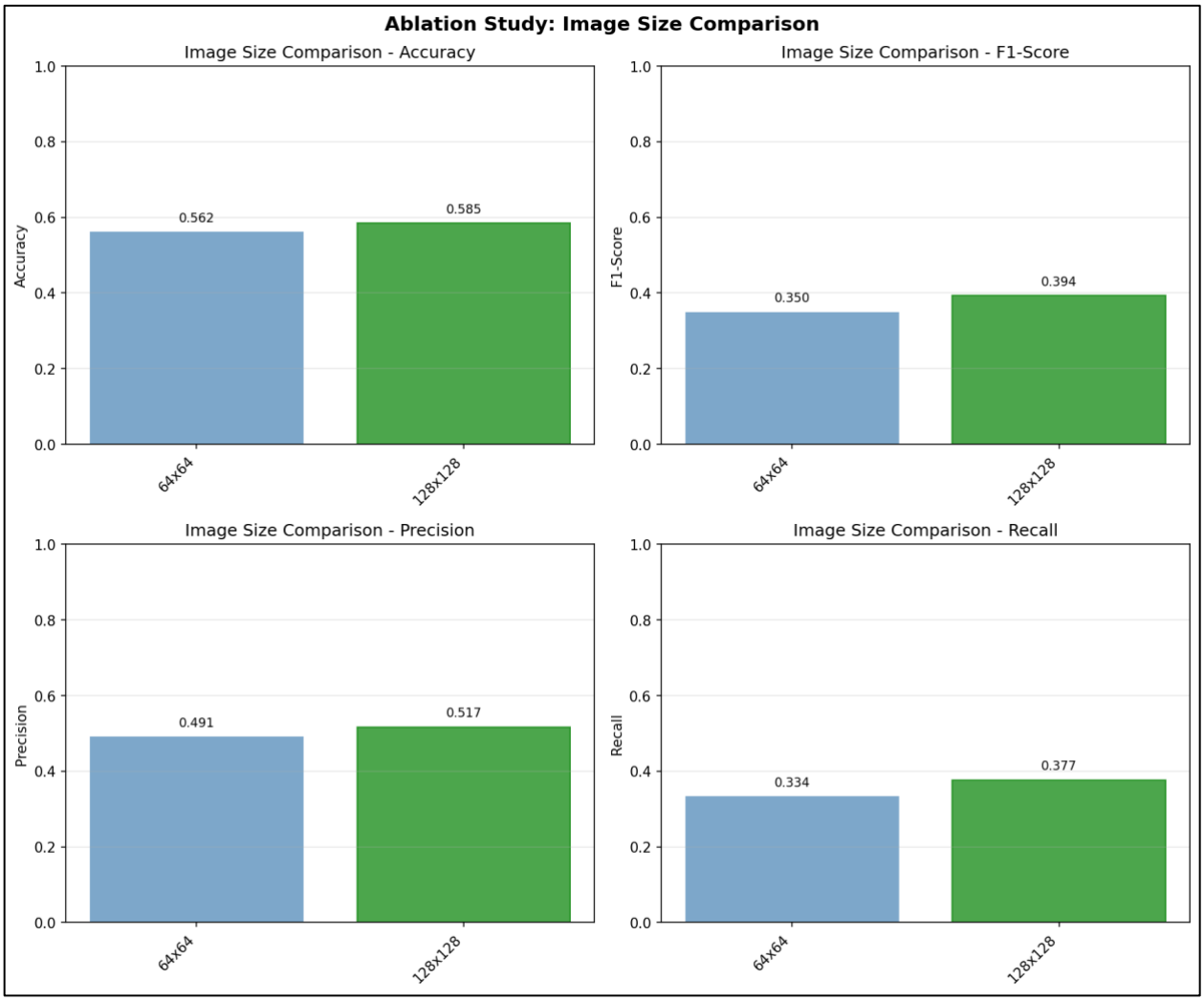

Figure 23: Impact of input image resolution on Random Forest performance. Higher resolution provides modest but consistent improvements.

Figure 24: 128x128 Confusion Matrix

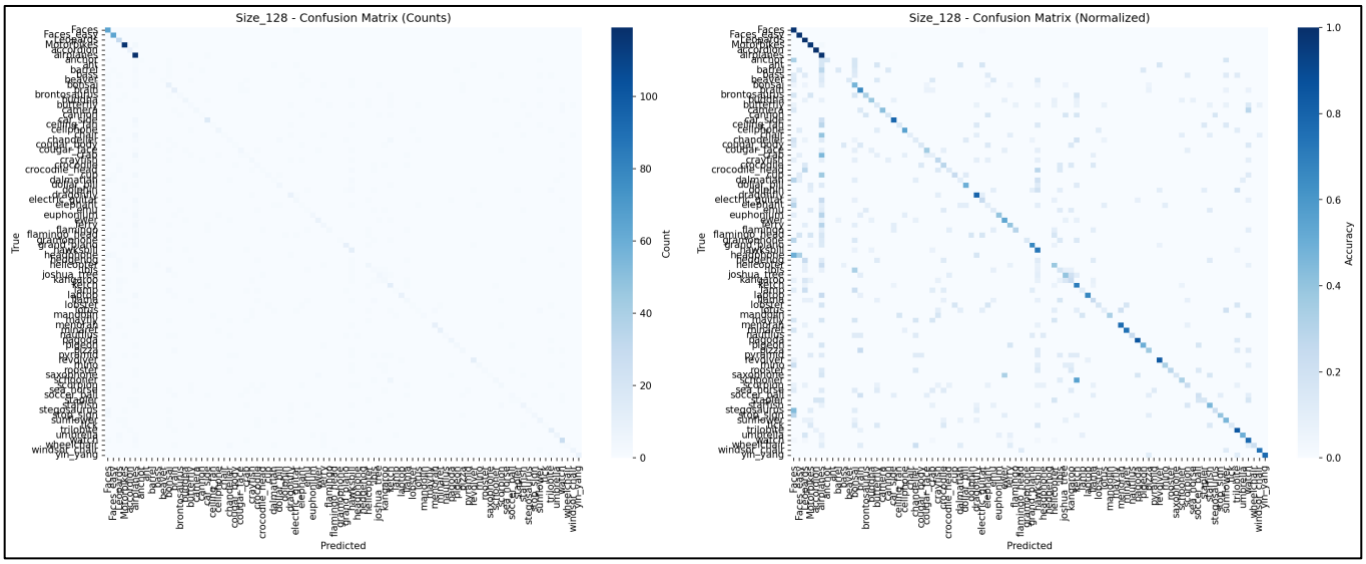

Figure 24: Confusion matrix for 128×128 input images.

Figure 25: 64x64 Confusion Matrix

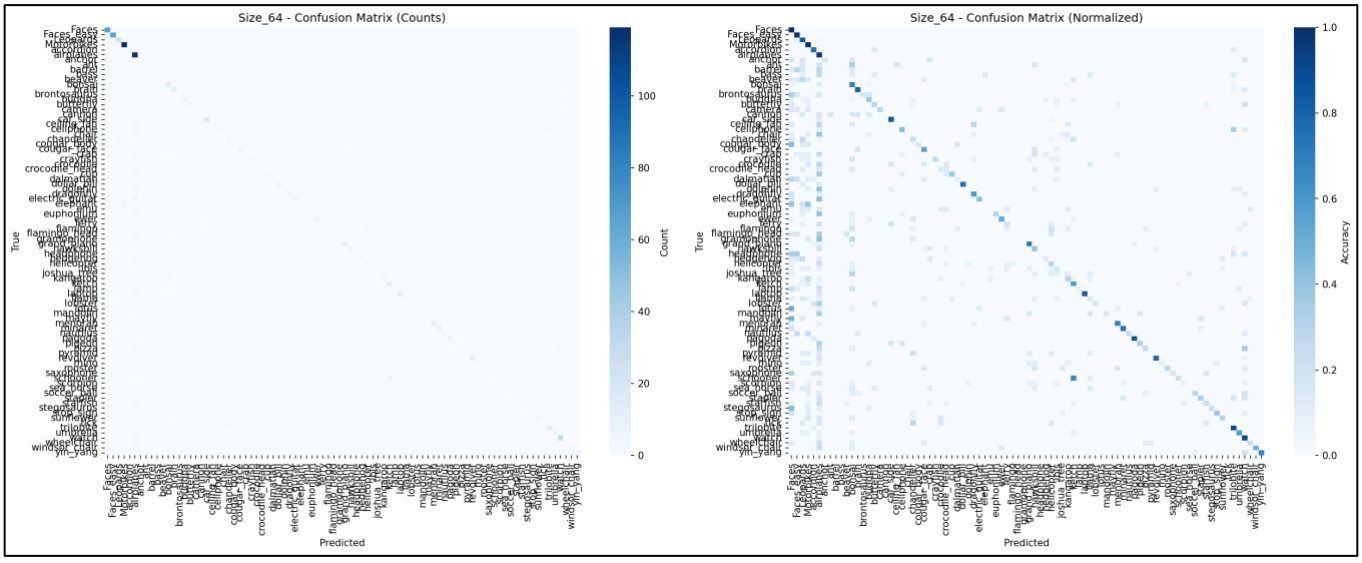

Figure 25: Confusion matrix for 64×64 input images. Lower resolution leads to more confusion, especially for detail-dependent categories.

3. 5 Summary

Ranking by Test Accuracy:
EfficientNet-B0: 94.52% > ResNet-18: 84.30% > SVM (HOG): 68.68% > KNN (HOG): 58.22% > Random Forest (HOG): 56.66%

Ablation Study:

- **Best augmentation strategy**: With augmentation (+1.39% accuracy)
- **Best optimizer**: Adam (+3.35% over SGD)
- **Best feature type**: HOG (+9.04% over raw pixels)
- **Best image size**: 128×128 (+2.29% over 64×64)

4. Discussion

The training results demonstrate a substantial performance gap between deep learning and classical machine learning. EfficientNet-B0 (94.52% accuracy) outperformed the best classical method, SVM (68.68%), by 25.84%. This improvement is driven by deep models' ability to automatically learn hierarchical feature representations, whereas classical models rely on rigid, hand-crafted HOG features. Furthermore, deep learning models benefited significantly from transfer learning using pretrained ImageNet weights, providing a massive advantage over classical methods that had to be trained from scratch on limited data.

Architectural efficiency also proved critical, as EfficientNet-B0 surpassed ResNet-18 by 10.22% despite having fewer parameters, this is achieved by compound scaling and optimized building blocks. Ablation studies further refined these insights, showing that the Adam optimizer and higher image resolutions consistently improved performance across metrics. While classical methods remain viable for extremely resource-constrained environments, the study concludes that deep learning with pretrained models is the optimal choice for high-accuracy image classification.

5. Conclusion

This comprehensive study compared classical machine learning and deep learning approaches for image classification on the Caltech-101 dataset. For image classification tasks today, deep learning with pretrained models should be the default choice if resources allow. EfficientNet provide a balance of accuracy and efficiency. Classical machine learning methods remain viable only if there is limited computational resources or when it is needed to interpret predictions and results. For this Caltech-101 dataset and similar multi-class object recognition tasks, EfficientNet-B0 with Adam optimizer is suggested, with standard data augmentation, and 224×224 input resolution as the optimal configuration among tested approaches.